

# KAVACH AI

## India's AI-Powered Digital Safety Shield

Defeating Counterfeiting, Fraud & Digital Arrest Scams

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Track: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement

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# 1. Problem Statement & Executive Context

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## The Growing Public Safety Crisis:

India registered over **1.14 million cybercrime complaints** in 2023 — a 60% surge from 2022. Among the most dangerous emerging threats are '**Digital Arrest**' scams, where victims are psychologically held hostage over video calls by criminals impersonating CBI, ED, or Police officers. The Ministry of Home Affairs reported citizens defrauded of over **Rs 1,776 crore** in the first nine months of 2024 alone.

Concurrently, counterfeit currency (Fake Indian Currency Notes — FICN) continues to undermine financial systems, with high-quality Rs 500 fake notes bypassing traditional manual detection. Existing law enforcement frameworks are reactive, stepping in only post-complaint rather than neutralizing threats at the point of contact.

## 2. Proposed Solution — KAVACH AI

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**KAVACH AI** is an integrated, multi-agent AI public safety platform designed to equip citizens, financial institutions, and law enforcement agencies (LEO) with real-time proactive tools to detect, disrupt, and neutralize digital fraud and counterfeit currency.

- **Citizen Fraud Shield:** Multilingual conversational advisory in 12 regional languages providing real-time call/message threat verdicts & 1-click reporting to NCRB 1930.
- **Digital Arrest Scam Detector:** ML-powered transcript & speech classifier trained on 15 call categories to detect impersonation, fear tactics, and urgency indicators.
- **Counterfeit Currency Agent:** Computer vision forensics analyzing microprint, intaglio edge sharpness, HSV color profiles, and printer frequency artifacts across denominations.
- **Fraud Network Graph AI:** Graph algorithms (Louvain Community Detection, PageRank) mapping money mule clusters, suspect nodes, and organized campaign infrastructure.
- **Geospatial Crime Hotspots:** Spatial analytics (DBSCAN & Kernel Density Estimation) identifying cybercrime hotspots and guiding police patrol prioritization.
- **LEO Command & Court Evidence:** End-to-end case management generating court-admissible PDF intelligence packages.

### 3. Hero Feature Showcase — Distinguishing the 3 Dashboards

KAVACH AI provides **three distinct, role-tailored hero dashboards** designed for Citizens, Law Enforcement Officers (LEO), and System Administrators. Each dashboard serves a unique operational purpose:

| Dashboard Role                                                      | Target Audience & Operational Goal                                                        | Distinguishing Hero Features & Capabilities                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CITIZEN DASHBOARD</b><br><a href="#">Public Safety Shield</a>    | Citizens, families, and banking customers verifying suspicious calls, messages, or notes. | <ul style="list-style-type: none"><li>• <b>Scam Detector:</b> Text &amp; Call Transcript classifier.</li><li>• <b>Currency Verification:</b> 6-layer CV note checks.</li><li>• <b>Media Forensics:</b> Deepfake face &amp; voice cloning.</li><li>• <b>RAG Advisory:</b> 12 Regional Indian languages.</li><li>• <b>1-Click NCRB 1930:</b> Direct official cybercrime reporting.</li></ul> |
| <b>LEO DASHBOARD</b><br><a href="#">Command &amp; Investigation</a> | Law Enforcement Officers, Cyber Cells, and Financial Investigators.                       | <ul style="list-style-type: none"><li>• <b>Case Management:</b> Lifecycle status &amp; suspect vault.</li><li>• <b>Fraud Network Graph:</b> Louvain mule ring clustering.</li><li>• <b>Geospatial Hotspots:</b> DBSCAN &amp; KDE density maps.</li><li>• <b>Court-Admissible Evidence:</b> Export PDF packages.</li></ul>                                                                  |
| <b>ADMIN DASHBOARD</b><br><a href="#">Platform Governance</a>       | System Administrators, MHA Authorities, & Network Operators.                              | <ul style="list-style-type: none"><li>• <b>User &amp; Role Control:</b> Elevate roles (Citizen→LEO).</li><li>• <b>Emergency Broadcast:</b> Dispatch priority SMS/Alerts.</li><li>• <b>System Audit Log:</b> Track all API &amp; security events.</li><li>• <b>Health Diagnostics:</b> Real-time FastAPI &amp; DB status.</li></ul>                                                         |

#### Role Isolation & Security Controls:

Each dashboard is strictly isolated via JWT role-based access control (RBAC) middleware (`LEO-only`, `ADMIN-only`). Citizens have no visibility into police case files or system audit registries, ensuring data confidentiality and public safety integrity.

## 4. KAVACH AI Landing Page Visual Interface Showcase

The public portal serves as India's primary digital safety shield, providing instant access to scanning tools, threat advisories, and emergency cybercrime reporting.



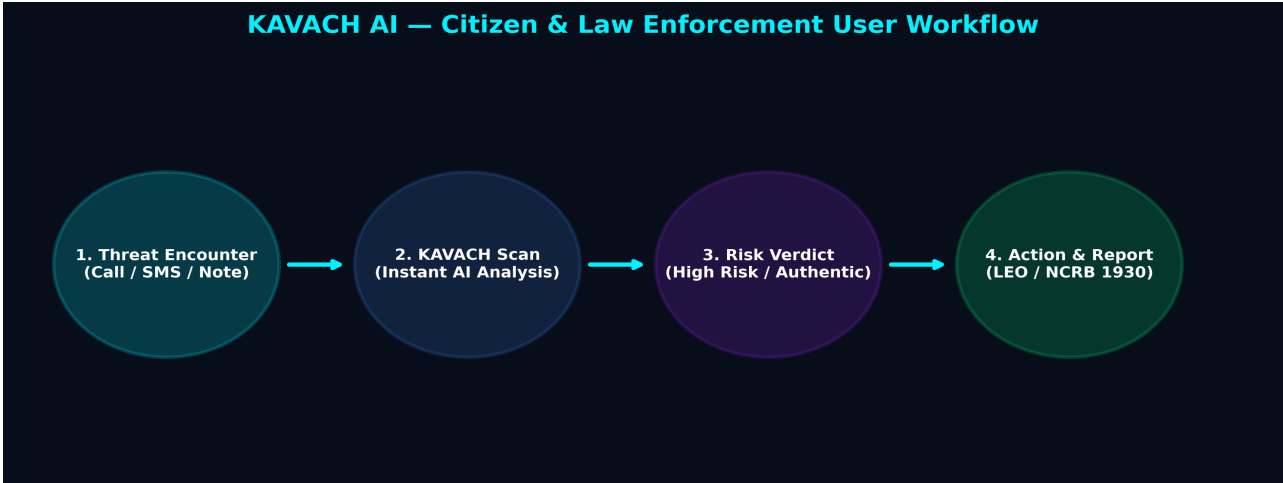
### Key Features Highlighted on the Landing Portal:

- **Hero Shield Section:** Immediate entry point for Citizens to scan suspicious call logs, text messages, and currency notes.
- **Interactive Feature Showcase:** Highlighting Digital Arrest protection, Counterfeit Currency verification, Fraud Ring mapping, and 12-language RAG advisories.
- **1-Click Helpline Access:** Instant emergency dispatch link to National Cyber Crime Reporting Portal (NCRB `cybercrime.gov.in`) and `1930` helpline.

## 5. Technology Stack & Technical Matrix

| Component       | Technologies Used                                     | Key Purpose                                  |
|-----------------|-------------------------------------------------------|----------------------------------------------|
| Frontend        | HTML5, Vanilla JS, TailwindCSS, Iconify, GSAP         | Responsive Citizen, LEO & Admin Dashboards   |
| Backend API     | FastAPI, Uvicorn, Python 3.11, Pydantic v2            | Async REST APIs & Microservices Layer        |
| AI & ML Models  | Scikit-Learn, NetworkX, SciPy, NumPy, Joblib, Whisper | Scam NLP, Call ML, Graph AI, Geospatial      |
| Computer Vision | OpenCV, Pillow (PIL), FFT Frequency Analysis          | Currency Note Forensics & Deepfake Detection |
| Database & Auth | SQLite, SQLAlchemy Async ORM, PyJWT, Bcrypt           | Persistent Data & Role-Based Access          |

### User Processing Journey Workflow:



## 6. Future Scope & Strategic Roadmap

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- **WebRTC Live Call Interception:** Integrating real-time audio stream monitoring directly on mobile devices to warn citizens mid-call during active digital arrest attempts.
- **Direct Carrier & WhatsApp Cloud API Integration:** Webhook bridges connecting telecom service providers and Meta WhatsApp Business API for instant threat blocking.
- **Distributed Graph & Database Scaling:** Upgrading SQLite persistence to enterprise PostgreSQL + Neo4j for multi-billion transaction graph analysis.
- **Hardware UV Sensor Integration:** Calibrating computer vision models for specialized UV/IR camera hardware on banking note counting machines.

## 7. Socio-Economic Impact & Evaluation Alignment

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KAVACH AI addresses the core evaluation criteria of the ET AI Hackathon 2026:

- **Innovation (25%):** Multi-agent fusion uniting NLP, Computer Vision, Graph AI, and Geospatial analytics into a single public safety shield.
- **Business & Social Impact (25%):** Safeguarding citizen life savings against psychological digital arrests and preventing counterfeit currency proliferation.
- **Technical Excellence (20%):** Self-contained, zero-external-dependency production architecture with 25+ REST endpoints and trained ML models.
- **Scalability & UX (30%):** Async FastAPI backend, 12 regional languages, and intuitive dark-mode glassmorphism interface.

### Conclusion:

Project **KAVACH AI** by Team **BitBuddies** provides a production-grade, multi-agent AI safety shield that shifts public safety from reactive post-crime reports to real-time predictive threat neutralization for citizens, law enforcement, and financial institutions.